

Loop: Loops are used to execute a block of code repeatedly until a condition is met or all items in a sequence are processed.

Types of Loop:

For Loop

While Loop

```
In [ ]: # For Loop  
#For loops is used to iterate over a sequence such as a list, tuple, string or range. It executes
```

```
In [2]: name = 'nit'  
for i in name:  
    print(i)    # It prints in sequence.
```

n
i
t

```
In [4]: name1 = [1,2,3,'hallo']  
  
for i in name1:  
    print(i)
```

1
2
3
hallo

```
In [5]: for i in [2,3,4,5,'hi']:  
        print(i)
```

2
3
4
5
hi

```
In [ ]: # range: The range() function is used to generate a sequence of numbers.  
# It is commonly used with a for loop when you want to repeat something a certain number of times
```

```
In [6]: for i in range (5):  
        print(i)
```

0
1
2
3
4

```
In [7]: for i in range (2,5):  
        print(i)
```

2
3
4

```
In [8]: for i in range(1,10,3):  
        print(i)
```

1
4
7

print the value which is divisible by 5

```
In [9]: for i in range(1,51):  
        if i%5 == 0:  
            print(i)
```

5
10
15
20
25
30
35
40
45
50

print the value which is divisible by 5 i dont want that value

```
In [10]: for i in range(1,51):  
         if i%5 != 0:  
             print(i)    # Here 5 has been skipped.
```

1
2
3
4
6
7
8
9
11
12
13
14
16
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24
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48
49

```
In [ ]: # LETS DISCUSS ABOUT 3 KEYWORDS -- BREAK || CONTINUE || PASS
        # BREAK STATEMNT - if you apply break statment in a loop then it will end the loop
        # Pass = skips block of code( function, class etc)
        # Continue= skips 1 step/iteration during loop
        # Break= jumps out of the function/loop
```

```
In [ ]: # write the code user ask chocklet from vendor machne write the basic code

ava = 5 # the machine as only 5 chocolates
x = int(input('Enter the number of chocolate you want: ?'))

i = 1
while i<=x:
    print(chocolate)
    i +=1

# if you check the user wants 10 choclets but availabe choclet is 5 but we got output as 10
# in this code we just declare but we dint apply any condition to it
```

Break: Break means "Stop the loop completely."

When Python sees break, it immediately exits the loop.

```
In [17]: available_chocolate = 5 # the machine as only 5 chocolates
x = int(input('Enter the number of chocolate you want: ?'))

i = 1
while i<=x:
    if i > available_chocolate: # we stop the execution but which code execution not entire
        break # break is statement | means jump out of the loop
    print('chocolate')
    i +=1
print('Buy for now')
```

chocolate
chocolate
chocolate
chocolate
chocolate
Buy for now

```
In [18]: for i in range (1,10):
        print(i)
```

1
2
3
4
5
6
7
8
9

```
In [23]: # i dont want 11 number i want only 5 number for the range of 1 to 10
```

```
for i in range(1,11):
    if i ==6:
        break
    print(i)
```

1
2
3
4
5

Continue: Continue means "Skip this item and move to the next one." The loop does not stop.

```
In [25]: for i in range(1,11):
        if i==3:
            continue
        print(i)
```

```
1
2
4
5
6
7
8
9
10
```

```
In [27]: for i in range(1,11):
         if i == 6:
             continue
         print('hello: ', i)
```

```
hello: 1
hello: 2
hello: 3
hello: 4
hello: 5
hello: 7
hello: 8
hello: 9
hello: 10
```

```
In [ ]: # Pass: Pass means "Do nothing." It is used when Python expects some code, but you don't want
```

```
In [28]: for i in range(1,11):
```

```
Cell In[28], line 1
    for i in range(1,11):
                        ^
_IncompleteInputError: incomplete input
```

```
In [31]: for i in range(1,11):
         pass
```

Patterns: Patterns are programs that print shapes using loops (for loops). They are commonly asked in interviews and coding practice.

```
In [32]: print('# # # #')
         print('# # # #')
         print('# # # #')
         print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [34]: for i in range(1,5):
         i = i+1
         print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [35]: for i in range (1,5):
         if i<5:
             print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [37]: for i in range(4):  
         print('#')
```

```
#  
#  
#  
#
```

```
In [38]: for i in range(4):  
         print('# # # #')
```

```
# # # #  
# # # #  
# # # #  
# # # #
```

```
In [40]: for j in range (4):  
         print('#', end = ' ')
```

```
# # # #
```

```
In [42]: for i in range(1,5):  
         print('# ' * i) # Decremental
```

```
#  
# #  
# # #  
# # # #
```

```
In [44]: for i in range(1,5):  
         for j in range (4):  
             if i>j:  
                 print('#', end = ' ')  
         print()
```

```
#
```

```
#  
#
```

```
#  
#  
#
```

```
#  
#  
#  
#
```